

K-디지털 트레이닝

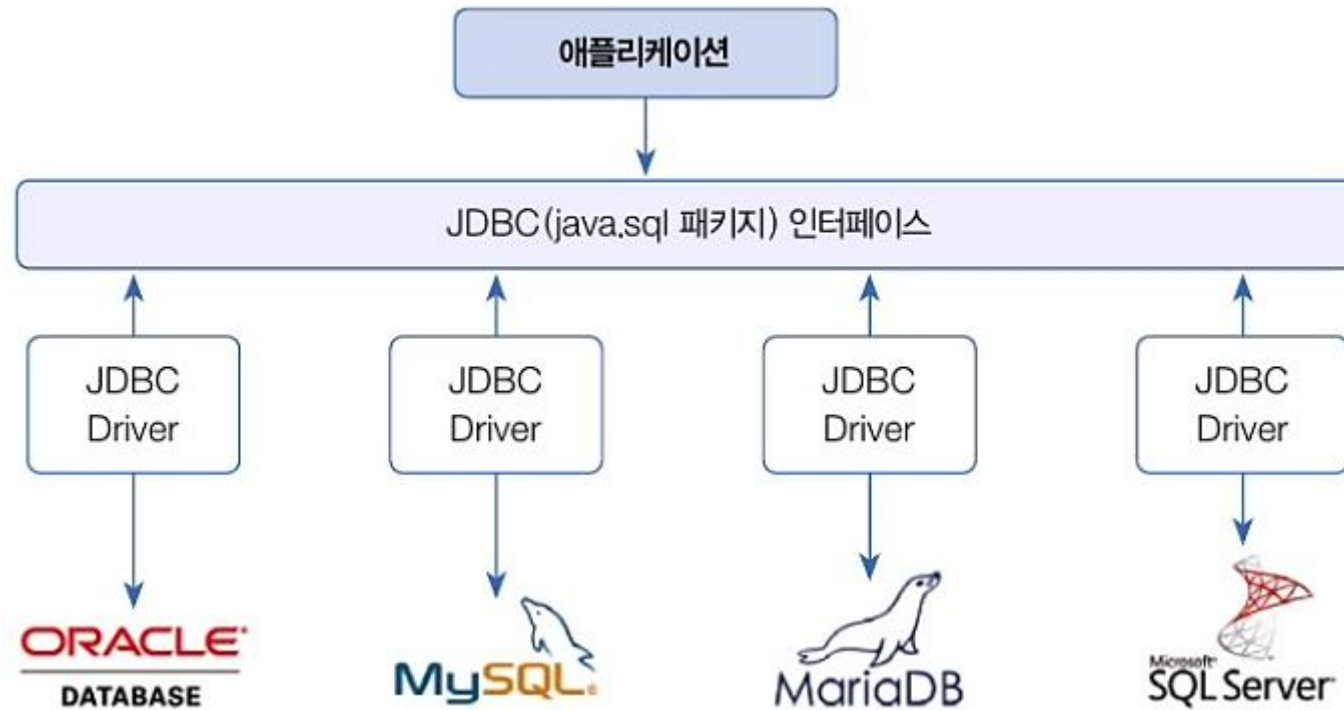
JDBC 프로그래밍

[KB] IT's Your Life

1 JDBC 프로그래밍

✓ JDBC Java Database Connectivity

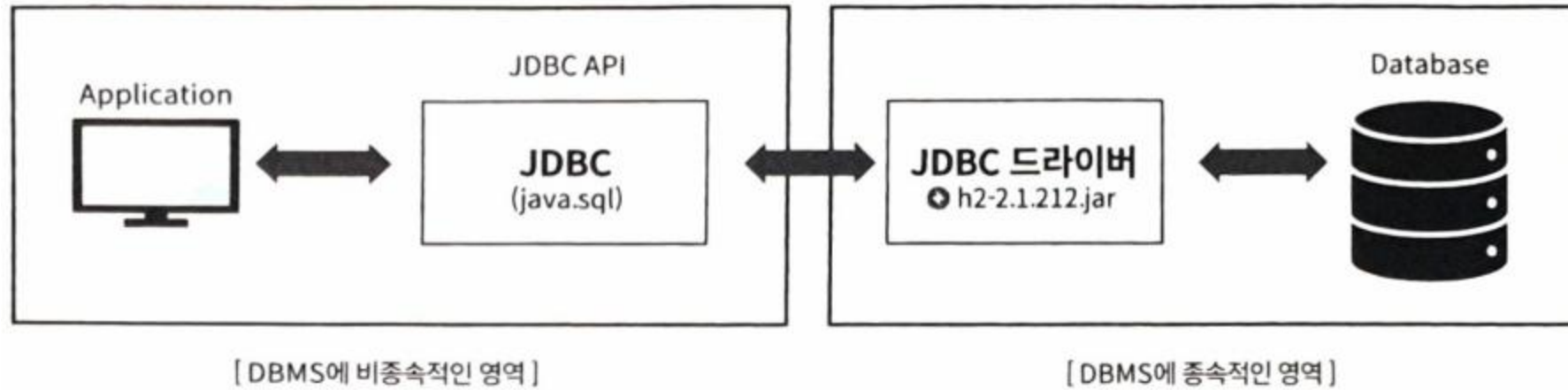
- 데이터베이스와 연결해서 입출력을 지원
- 데이터베이스 관리시스템(DBMS)의 종류와 상관없이 동일하게 사용할 수 있는 클래스와 인터페이스로 구성



1 JDBC 프로그래밍

✓ JDBC

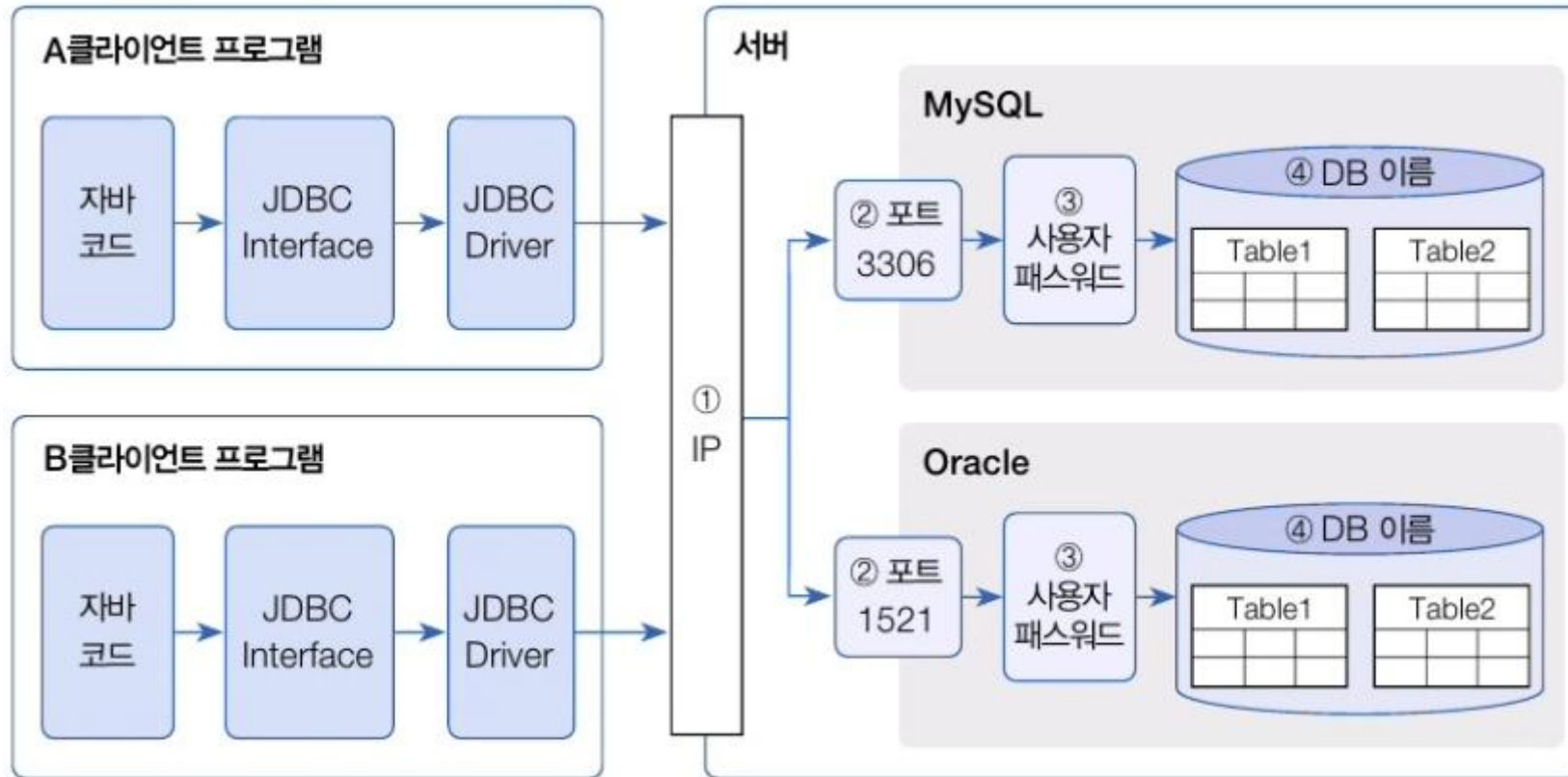
○ JDBC 개념



1 JDBC 프로그래밍

✓ JDBC

○ JDBC 개념



1 JDBC 프로그래밍

✓ 데이터베이스 준비

```
CREATE DATABASE jdbc_ex;
```

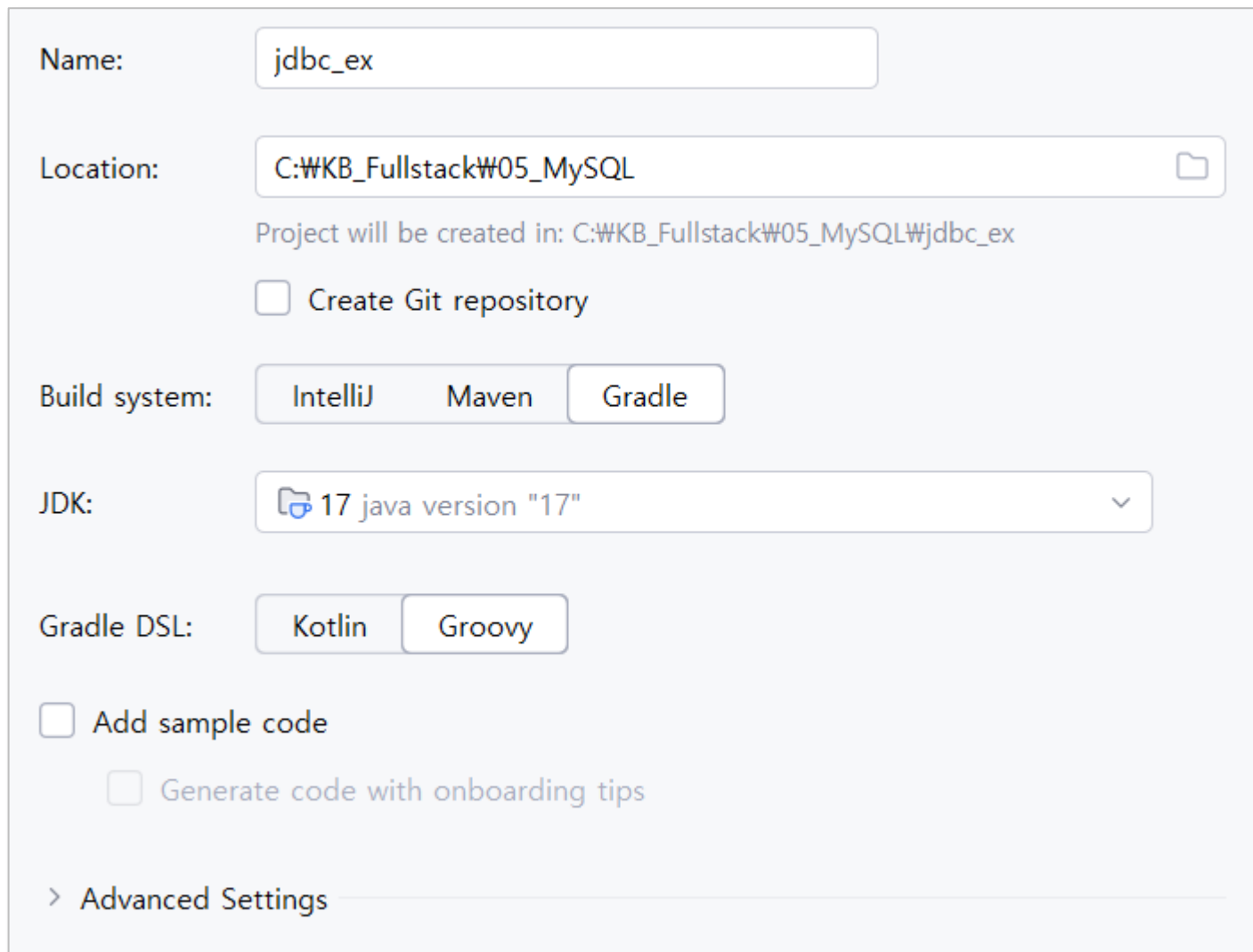
✓ 사용자 준비

```
CREATE USER 'scoula'@'%' IDENTIFIED BY '1234';  
GRANT ALL PRIVILEGES ON jdbc_ex.* TO 'scoula'@'%';  
FLUSH PRIVILEGES;
```

1 JDBC 프로그래밍

✓ 프로젝트 생성

- Name: jdbc_ex
- Build System : gradle
- Group Id: org.scoula



The image shows the 'New Project' dialog in IntelliJ IDEA. The 'Name' field is set to 'jdbc_ex'. The 'Location' field is set to 'C:\WKB_Fullstack\W05_MySQL', with a subtext indicating the project will be created in 'C:\WKB_Fullstack\W05_MySQL\jdbc_ex'. The 'Build system' is set to 'Gradle' (with 'IntelliJ' and 'Maven' as alternatives). The 'JDK' is set to '17 java version "17"'. The 'Gradle DSL' is set to 'Kotlin' (with 'Groovy' as an alternative). There are checkboxes for 'Create Git repository' (unchecked), 'Add sample code' (unchecked), and 'Generate code with onboarding tips' (unchecked). An 'Advanced Settings' link is at the bottom.

Name: jdbc_ex

Location: C:\WKB_Fullstack\W05_MySQL
Project will be created in: C:\WKB_Fullstack\W05_MySQL\jdbc_ex

☐ Create Git repository

Build system: IntelliJ Maven **Gradle**

JDK: 17 java version "17"

Gradle DSL: Kotlin **Groovy**

☐ Add sample code
☐ Generate code with onboarding tips

> Advanced Settings

1 JDBC 프로그래밍

✓ MySQL Connector

- <https://mvnrepository.com/>
- MySQL 검색



1. MySQL Connector/J 585 usages

`com.mysql » mysql-connector-j`

MySQL Connector/J is a JDBC Type 4 driver, which means that it is pure Java implementation of the MySQL protocol and does not rely on the registration with the Driver Manager, standard support for large update counts, support for localized time zone, support for the java.time package, support for JDBC-4.x XML schema, support for the NCHAR, NVARCHAR, and CLOB data types.

Last Release on Jan 16, 2024

Version	Vulnerabilities	Repository	Usages	Date
8.3.x 8.3.0		Central	168	Jan 16, 2024
8.2.x 8.2.0		Central	127	Oct 25, 2023
8.1.x 8.1.0		Central	164	Jul 18, 2023
8.0.33		Central	279	Apr 18, 2023
8.0.x 8.0.32		Central	139	Jan 18, 2023
		Central	106	Oct 14, 2022

[Maven](#)
[Gradle](#)
[Gradle \(Short\)](#)
[Gradle \(Kotlin\)](#)
[SBT](#)
[Ivy](#)
[Grape](#)
[Leiningen](#)
[Buildr](#)

```
// https://mvnrepository.com/artifact/com.mysql/mysql-connector-j
implementation 'com.mysql:mysql-connector-j:8.3.0'
```

- Lombok도 추가

1 JDBC 프로그래밍

build.gradle

```
...
dependencies {
    implementation 'com.mysql:mysql-connector-j:8.3.0'
    compileOnly 'org.projectlombok:lombok:1.18.30'
    annotationProcessor 'org.projectlombok:lombok:1.18.30'

    testCompileOnly 'org.projectlombok:lombok:1.18.30'
    testAnnotationProcessor 'org.projectlombok:lombok:1.18.30'

    testImplementation platform('org.junit:junit-bom:5.9.1')
    testImplementation 'org.junit.jupiter:junit-jupiter'
}
...
```

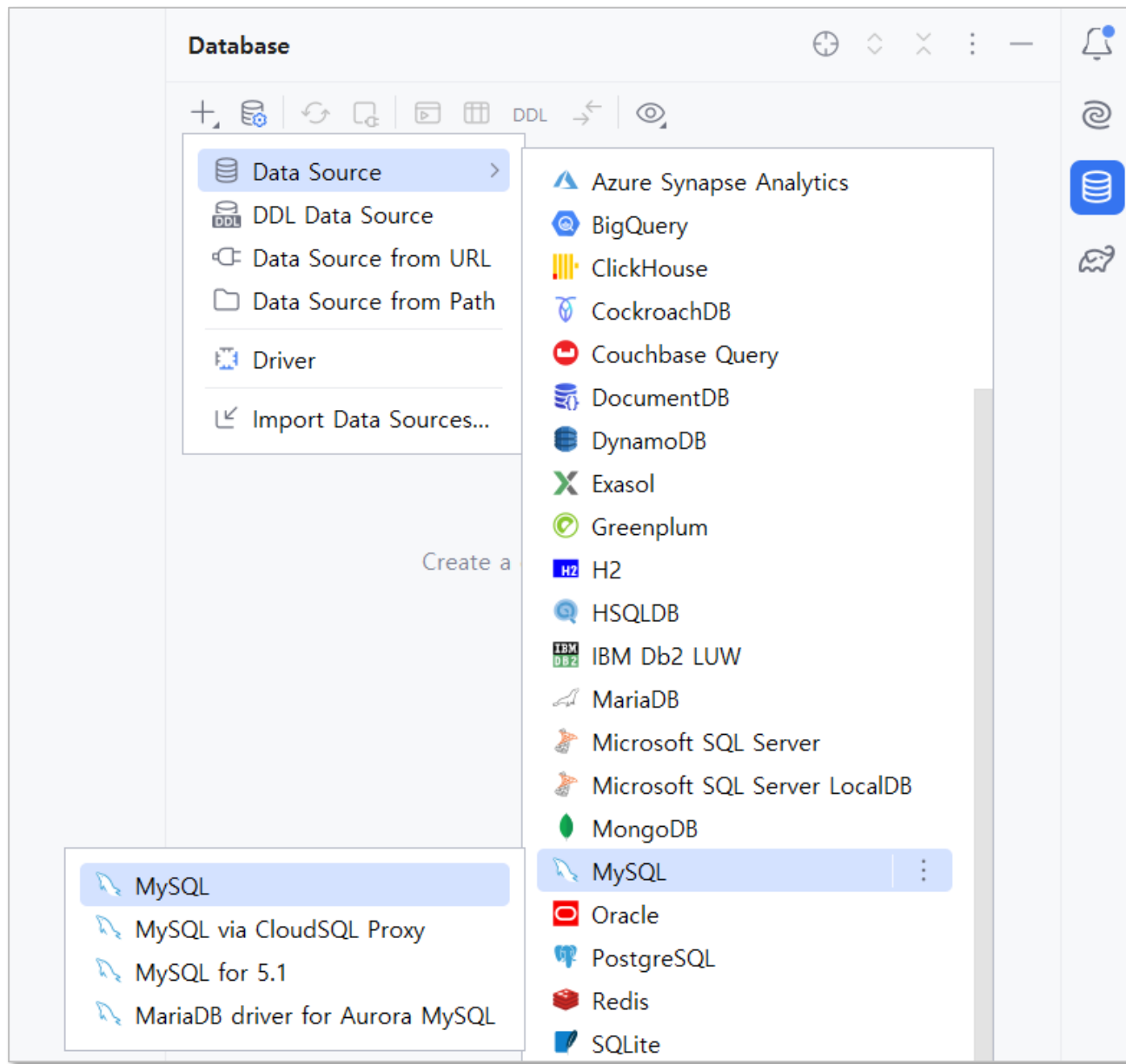
○ 수정 후 Sync 실행

○ 프로젝트 설정

- Annotation Processing 활성화

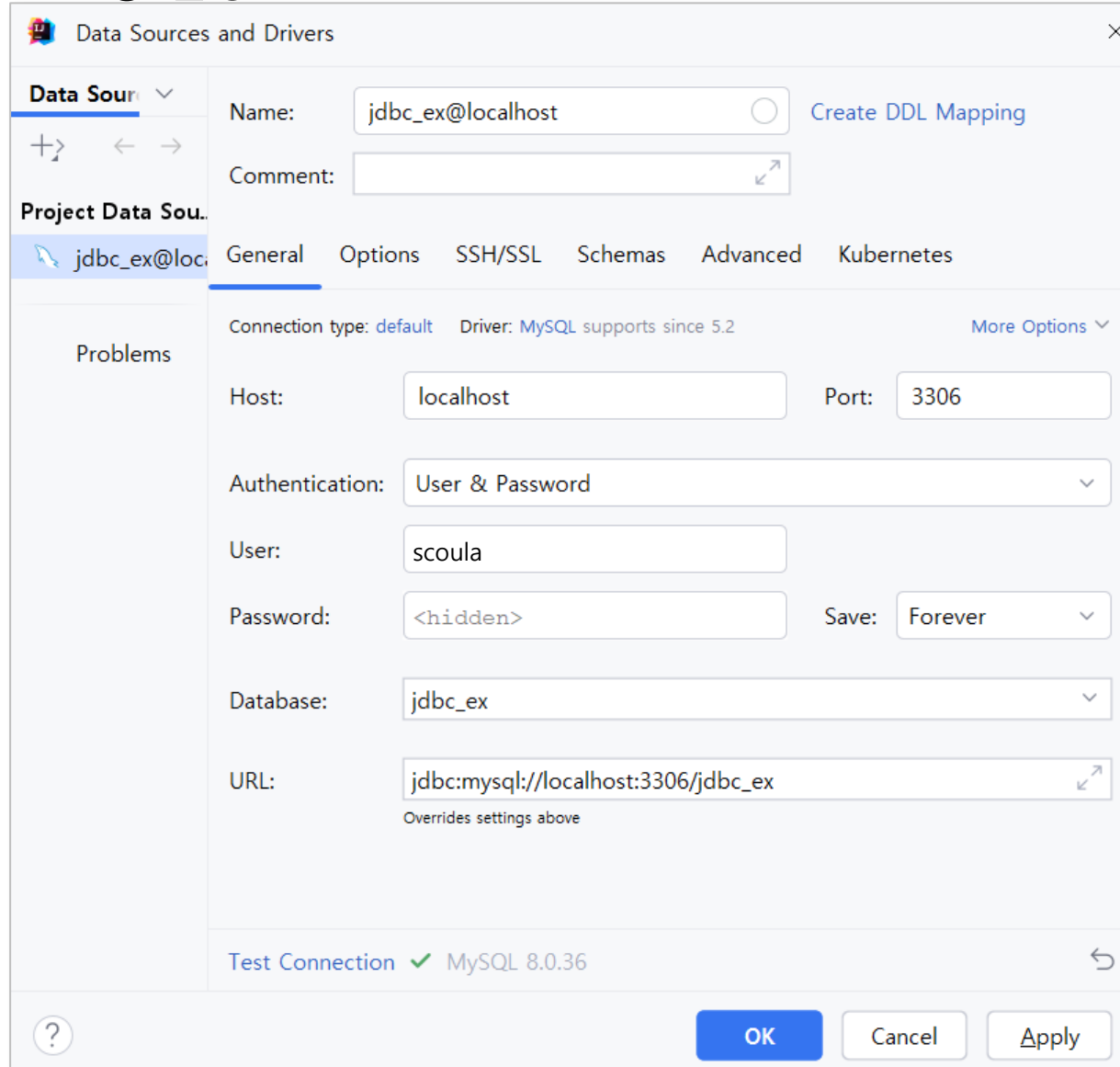
1 JDBC 프로그래밍

✓ IntelliJ Datasource 기능 설정



1 JDBC 프로그래밍

✓ IntelliJ Datasource 기능 설정



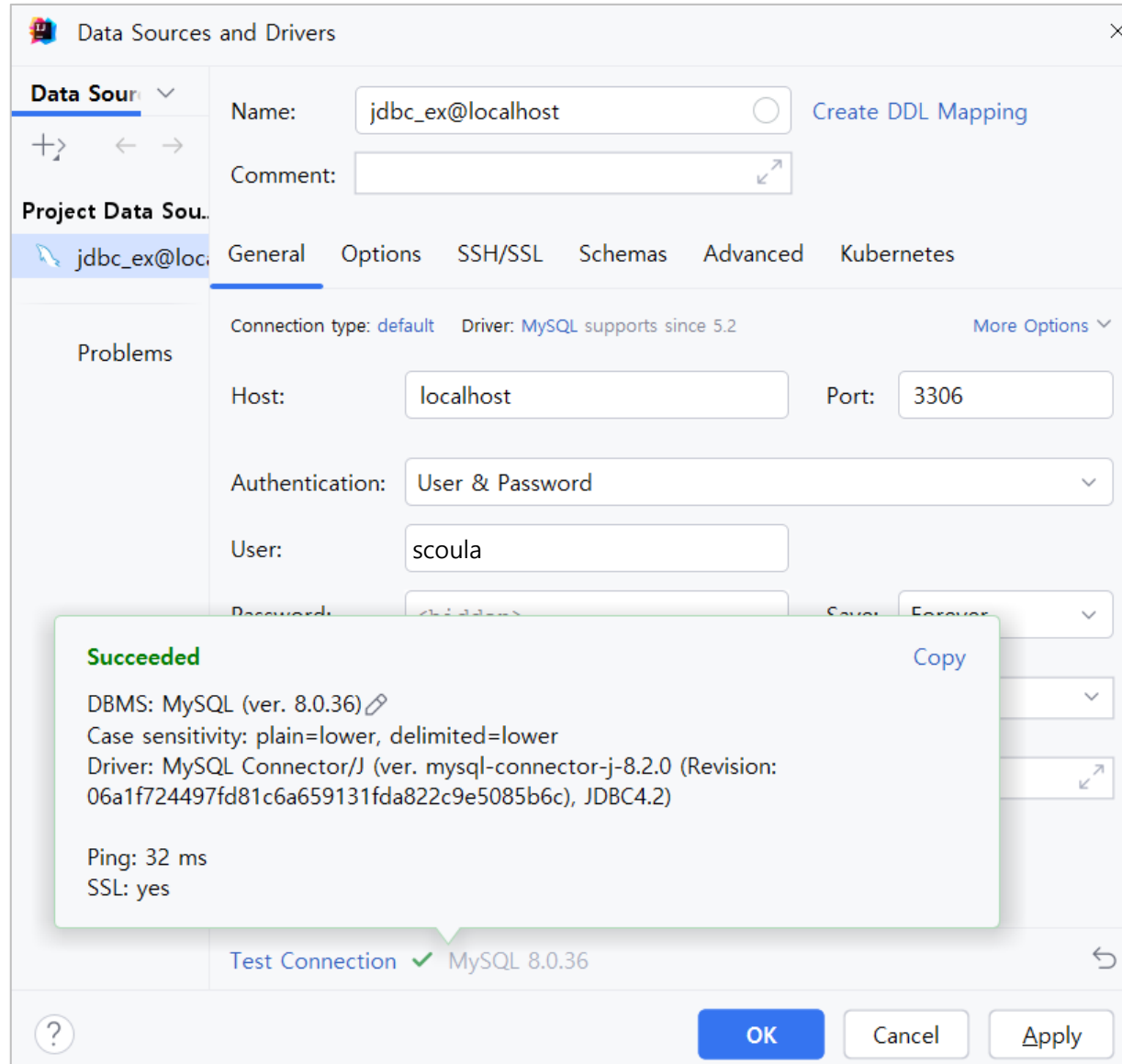
The image shows the 'Data Sources and Drivers' dialog box in IntelliJ IDEA. The 'Data Sources' tab is selected, and a new data source named 'jdbc_ex@localhost' is being configured. The 'General' tab is active, showing the following settings:

- Name: jdbc_ex@localhost
- Comment: (empty)
- Connection type: default
- Driver: MySQL supports since 5.2
- Host: localhost
- Port: 3306
- Authentication: User & Password
- User: scoula
- Password: <hidden>
- Save: Forever
- Database: jdbc_ex
- URL: jdbc:mysql://localhost:3306/jdbc_ex

The 'Test Connection' button is highlighted, and the status shows 'MySQL 8.0.36'.

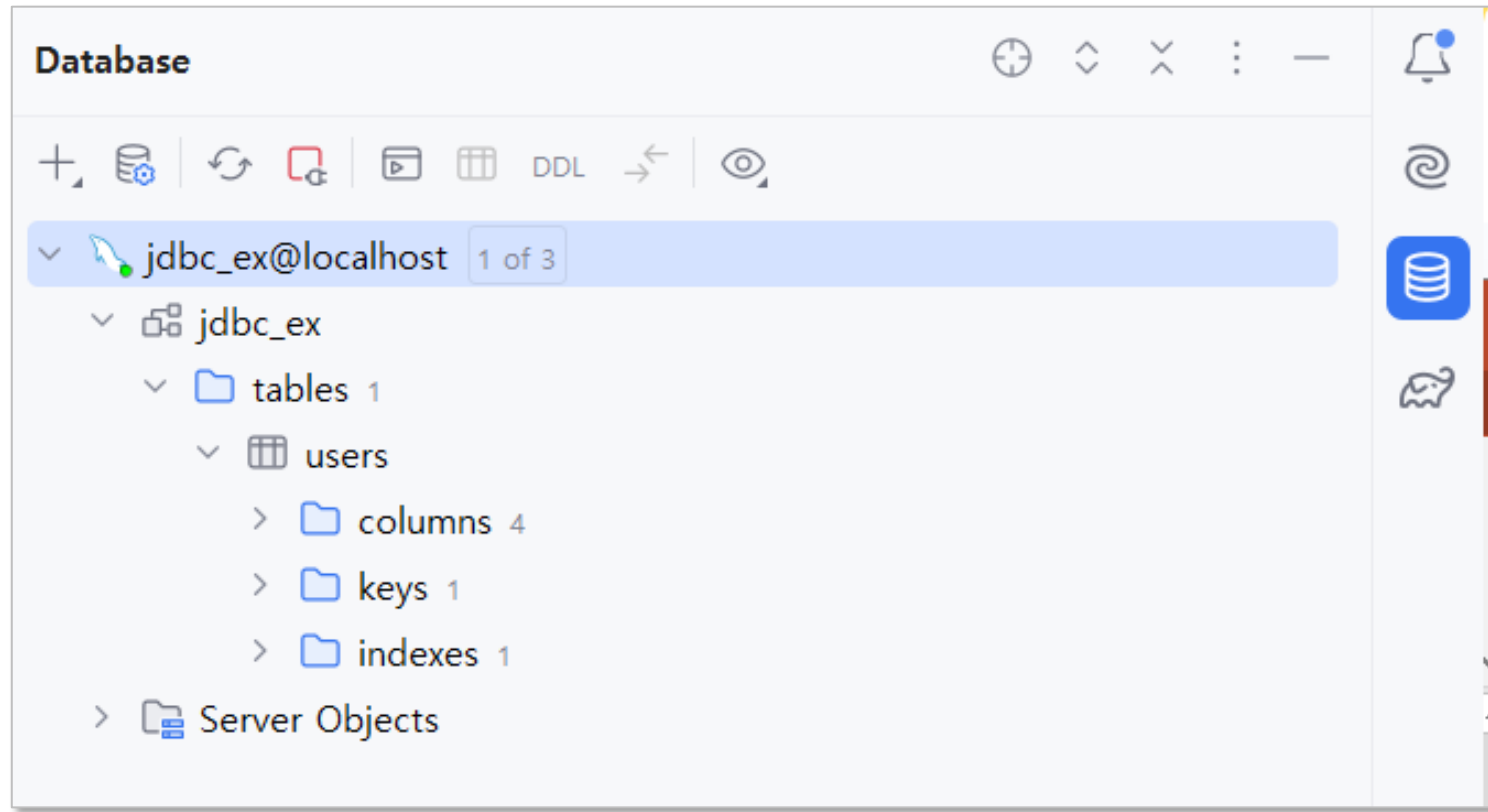
1 JDBC 프로그래밍

✓ IntelliJ Datasource 기능 설정



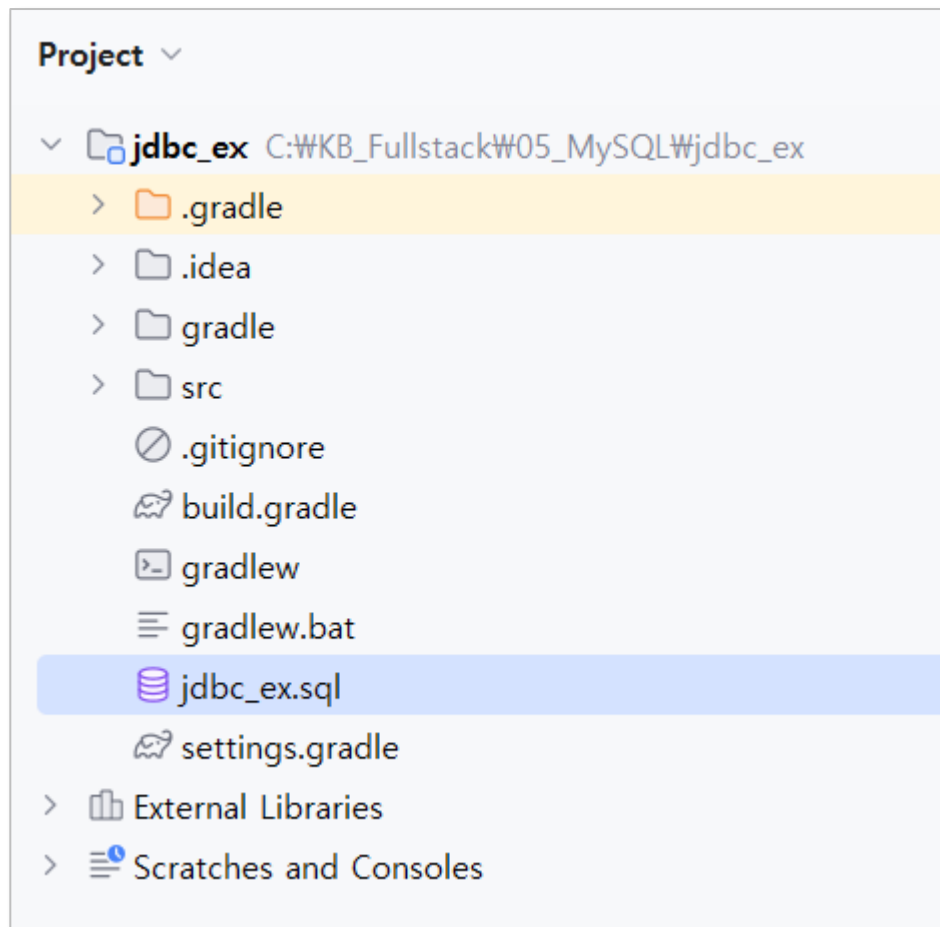
1 JDBC 프로그래밍

✔ datasource 목록



1 JDBC 프로그래밍

✓ sql 파일 만들기



1 JDBC 프로그래밍

✓ 쿼리 실행

1. datasource 선택

2. schema(database) 선택

4. 쿼리 실행

3. 쿼리 작성

select * from users;

Output jdbc_ex.users Tx

3 rows

ID	PASSWORD	NAME	ROLE	
1	admin	admin123	관리자	ADMIN
2	guest	guest123	방문자	USER
3	member	member123	일반회원	USER

1 JDBC 프로그래밍

✓ 데이터 준비

```
use jdbc_ex;
```

```
CREATE TABLE USERS (  
  ID VARCHAR(12) NOT NULL PRIMARY KEY,  
  PASSWORD VARCHAR(12) NOT NULL,  
  NAME VARCHAR(30) NOT NULL,  
  ROLE VARCHAR(6) NOT NULL  
);
```

```
INSERT INTO USERS(ID, PASSWORD, NAME, ROLE)  
VALUES('guest', 'guest123', '방문자', 'USER');
```

```
INSERT INTO USERS(ID, PASSWORD, NAME, ROLE)  
VALUES('admin', 'admin123', '관리자', 'ADMIN');
```

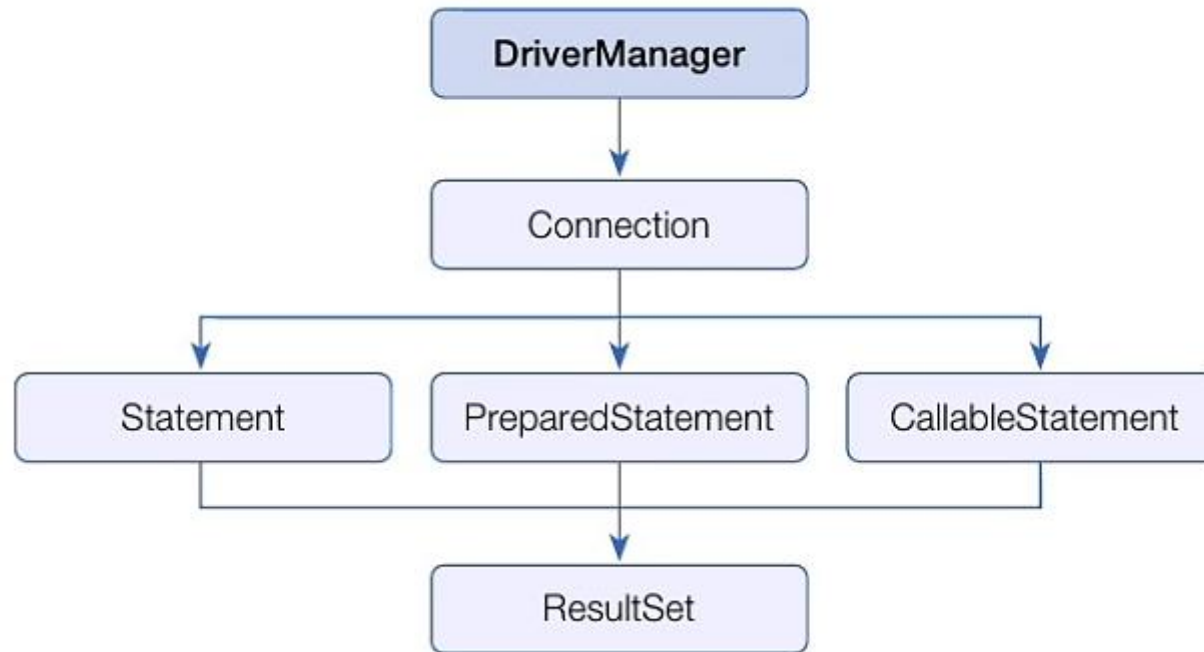
```
INSERT INTO USERS(ID, PASSWORD, NAME, ROLE)  
VALUES('member', 'member123', '일반회원', 'USER');
```

```
SELECT * FROM USERS;
```

1 JDBC 프로그래밍

✓ JDBC의 핵심 인터페이스/클래스

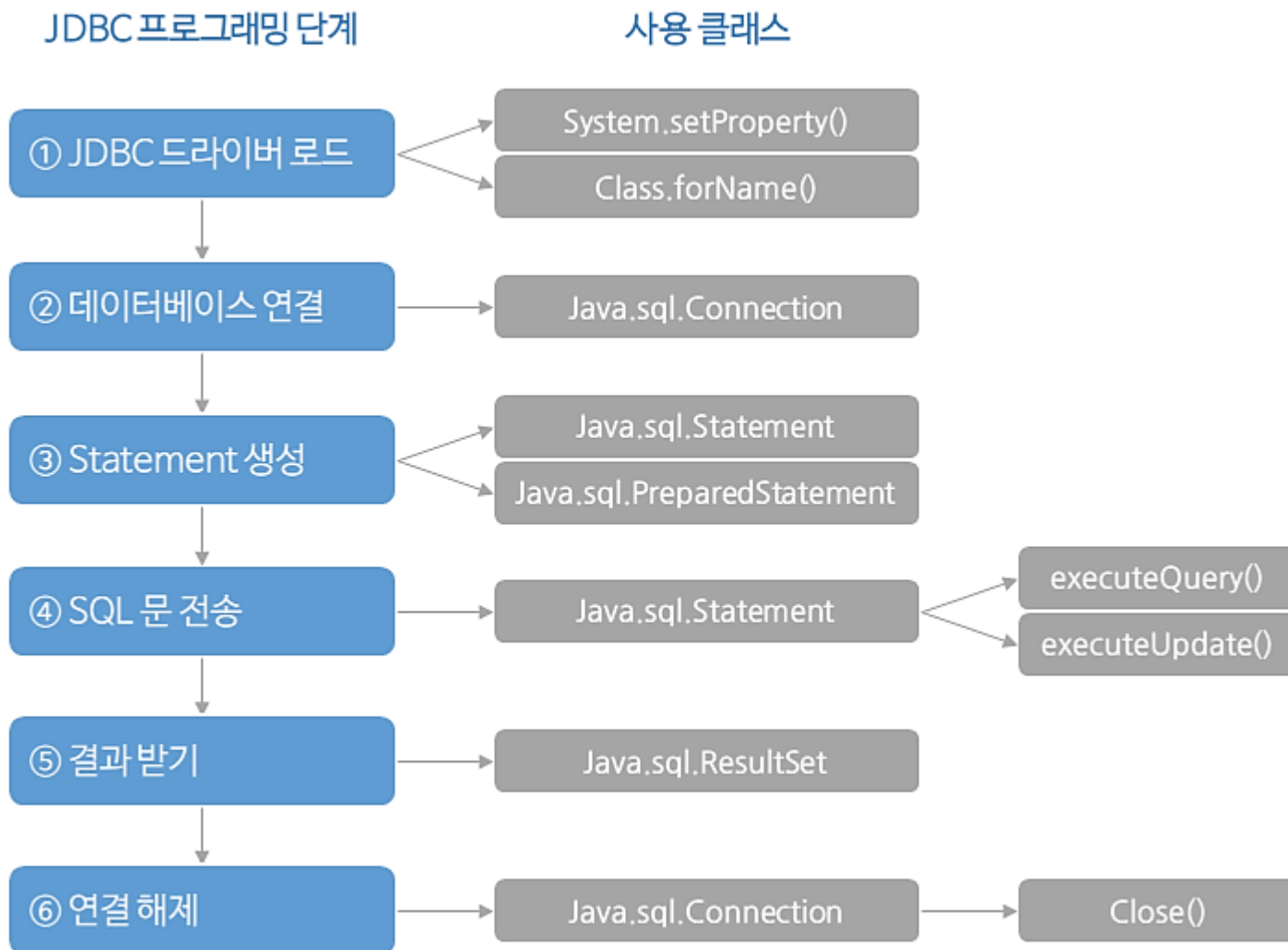
JDBC (java.sql 패키지)



1 JDBC 프로그래밍

✓ JDBC

○ JDBC 개발 절차



1 JDBC 프로그래밍

✓ DB 연결

○ 드라이버 확인

```
Class.forName("com.mysql.cj.jdbc.Driver");
```

→ 없으면 ClassNotFoundException 발생

○ Connection 객체

- 데이터베이스에 연결 세션을 만듦
- ```
Connection conn = DriverManager.getConnection("연결 문자열", "사용자", "비밀번호")
```
- 연결 문자열  

```
"jdbc:mysql://[host]:[포트]/[db이름]"
```

→ 

```
jdbc:mysql://127.0.0.1:3306/jdbc_ex
```
- ```
String url = "jdbc:mysql://127.0.0.1:3306/jdbc_ex";
```
- ```
Connection conn = DriverManager.getConnection(url, "jdbc_ex", "jdbc_ex");
```

## 1 JDBC 프로그래밍

 ConnectionTest.java

```
package org.scoula.jdbc_ex.test;

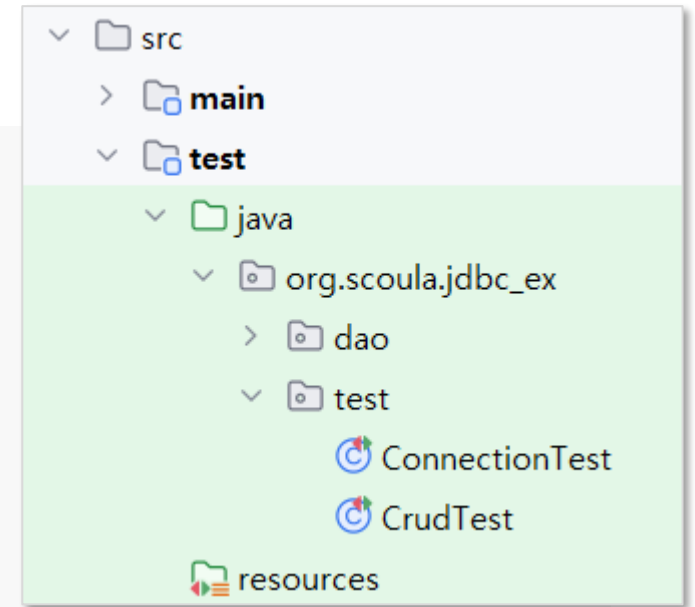
import org.junit.jupiter.api.DisplayName;
import org.junit.jupiter.api.Test;
import org.scoula.jdbc_ex.common.JDBCUtil;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class ConnectionTest {

 @Test
 @DisplayName("jdbc_ex 데이터베이스에 접속한다.")
 public void testConnection() throws SQLException, ClassNotFoundException {
 Class.forName("com.mysql.cj.jdbc.Driver");
 String url = "jdbc:mysql://127.0.0.1:3306/jdbc_ex";
 String id = "scoula";
 String password = "1234";
 Connection conn = DriverManager.getConnection(url, id, password);
 System.out.println("DB 연결 성공");
 conn.close();
 }
}
```

DB 연결 성공



# 1 JDBC 프로그래밍

## ✓ 모듈화해야 할 코드

- 데이터베이스 연결 및 닫기 작업은 항상 필요함

→ `common.JDBCUtil`

# 1 JDBC 프로그래밍

## resources::/application.properties

```
driver=com.mysql.cj.jdbc.Driver
url=jdbc:mysql://127.0.0.1:3306/jdbc_ex
id=scoula
password=1234
```

# 1 JDBC 프로그래밍

## JDBCUtil.java

```
package org.scoula.jdbc_ex.common;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;
import java.util.Properties;

public class JDBCUtil {
 static Connection conn = null;
 static {
 try {
 Properties properties = new Properties();
 properties.load(JDBCUtil.class.getResourceAsStream("/application.properties"));
 String driver = properties.getProperty("driver");
 String url = properties.getProperty("url");
 String id = properties.getProperty("id");
 String password = properties.getProperty("password");

 Class.forName(driver);
 conn = DriverManager.getConnection(url, id, password);
 } catch (Exception e) {
 e.printStackTrace();
 }
 }
}
```

# 1 JDBC 프로그래밍

## JDBCUtil.java

```
public static Connection getConnection() {
 return conn;
}

public static void close() {
 try {
 if (conn != null) {
 conn.close();
 conn = null;
 }
 } catch (SQLException e) {
 e.printStackTrace();
 }
}
}
```

# 1 JDBC 프로그래밍

## ConnectionTest.java

```
package org.scoula.jdbc_ex.test;
...

public class ConnectionTest {

 ...
 @Test
 @DisplayName("jdbc_ex에 접속한다.(자동 닫기)")
 public void testConnection2() throws SQLException {
 try(Connection conn = JDBCUtil.getConnection()) {
 System.out.println("DB 연결 성공");
 }
 }
}
```



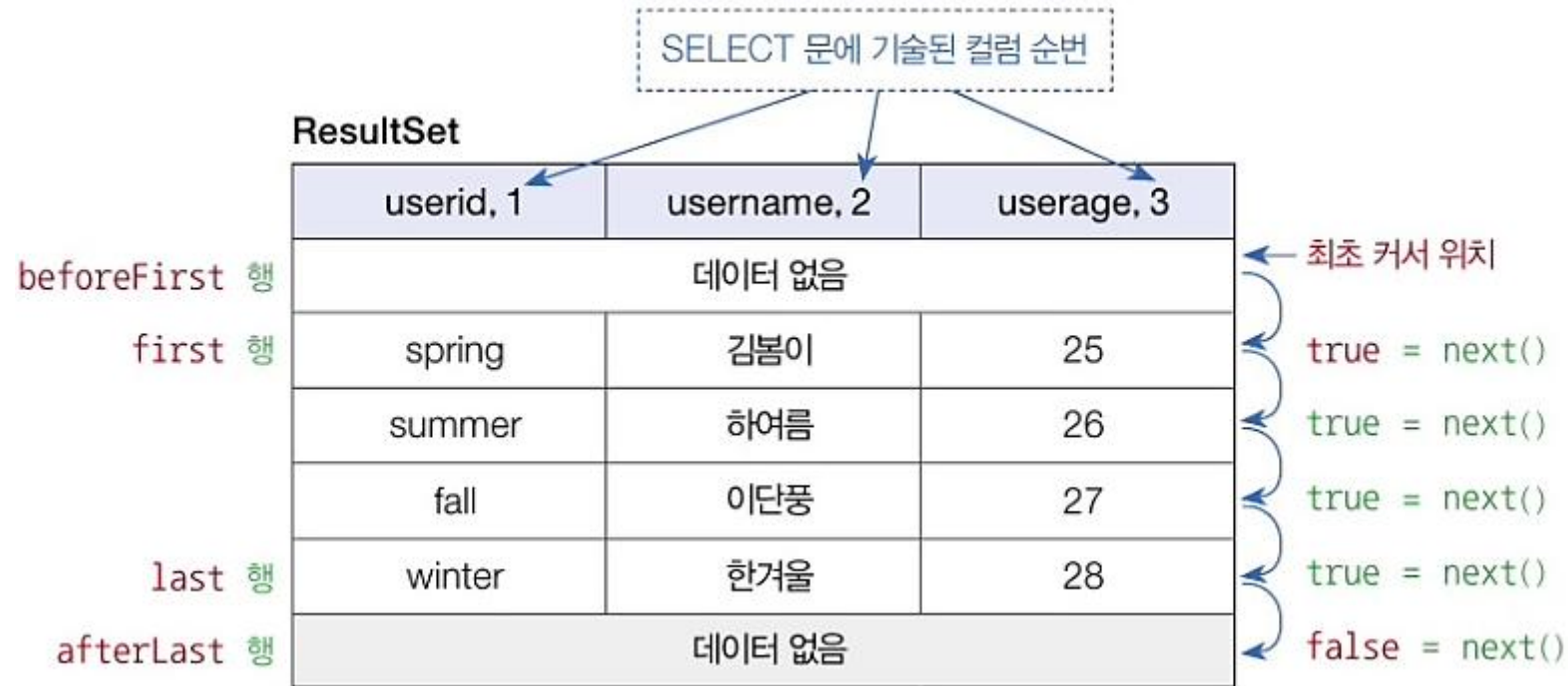
# 1 JDBC 프로그래밍

## ✓ Statement

- SQL 문 실행 클래스
- Connection 객체를 통해 생성  
Statement stmt = conn.createStatement();
- SQL 실행 메서드
  - ResultSet executeQuery(SQL문) : select문 실행
  - int executeUpdate(SQL문) : insert, update, delete 문 실행

## 1 JDBC 프로그래밍

## ✔ ResultSet



### ○ 컬럼 값 추출

- getXxxx("컬러명")
  - Xxx : 추출하고자하는 데이터 타입명
  - getString(), getInt(), getLong(), getDouble()

## ✓ PreparedStatement

### ○ SQL문에 값을 넣을 때 파라미터화 해서 처리

```
String sql = "INSERT INTO USERS(ID, PASSWORD, NAME, ROLE) " +
 "VALUES(?, ?, ?, ?)";
```

### ○ Connection 객체를 통해 생성

```
PreparedStatement pstmt = conn.prepareStatement(sql);
```

### ○ 파라미터 설정

- pstmt.setXxx(파라미터번호, 값)
  - setString(), setInt(), setLong(), setDouble()

### ○ SQL문 실행

```
int count = pstmt.executeUpdate();
```

# 1 JDBC 프로그래밍

## ✓ Statement로 Insert 문 실행하기

```
String sql = "INSERT INTO USERS(ID, PASSWORD, NAME, ROLE)" +
 "VALUES('member2', 'member123', '일반회원', 'USER')";
```

```
int count = stmt.executeUpdate(sql);
```

### ○ 값을 변수로 대체한다면 ?

```
String userId = "member2";
String password = "member123";
String name = "일반회원";
String role = "USER";
```

```
String sql = "INSERT INTO USERS(ID, PASSWORD, NAME, ROLE)" +
 "VALUES('" + userId + "', '" + password + "', '" +
 name + "', '" + role + "')";
```

→ PreparedStatement로 처리

# 1 JDBC 프로그래밍

## CrudTest.java

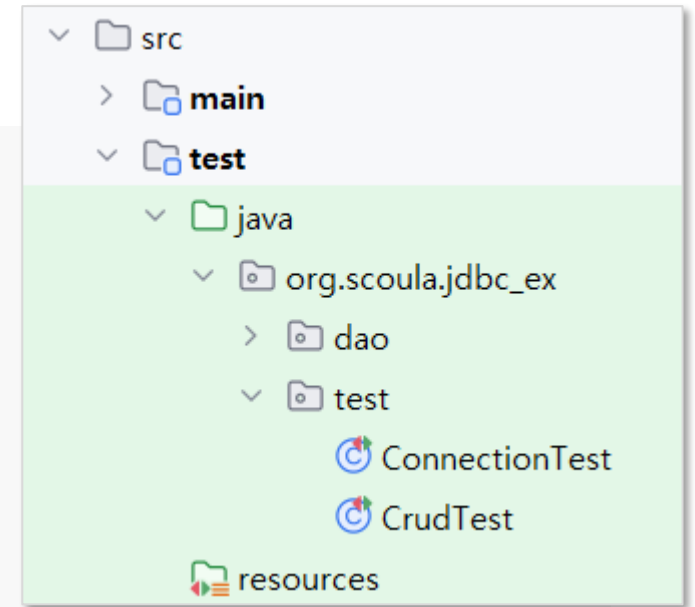
```
package org.scoula.jdbc_ex.test;

import org.junit.jupiter.api.*;
import org.scoula.jdbc_ex.common.JDBCUtil;

import java.sql.*;

@TestMethodOrder(MethodOrderer.OrderAnnotation.class)
public class CrudTest {
 Connection conn = JDBCUtil.getConnection();

 @AfterAll
 static void tearDown() {
 JDBCUtil.close();
 }
}
```



# 1 JDBC 프로그래밍

## CrudTest.java

```
@Test
@DisplayName("새로운 user를 등록한다.")
@Order(1)
public void insertUser() throws SQLException {
 String sql = "insert into users(id, password, name, role) values(?, ?, ?, ?)";
 try (PreparedStatement pstmt = conn.prepareStatement(sql)) {
 pstmt.setString(1, "scoula");
 pstmt.setString(2, "scoula3");
 pstmt.setString(3, "스콜라");
 pstmt.setString(4, "USER");

 int count = pstmt.executeUpdate();
 Assertions.assertEquals(1, count);
 }
}
```

# 1 JDBC 프로그래밍

## CrudTest.java

```
@Test
@DisplayName("user 목록을 추출한다.")
@Order(2)
public void selectUser() throws SQLException {
 String sql = "select * from users";
 try(Statement stmt = conn.createStatement();
 ResultSet rs = stmt.executeQuery(sql);
) {
 while(rs.next()) {
 System.out.println(rs.getString("name"));
 }
 }
}
```

# 1 JDBC 프로그래밍

## CrudTest.java

```
@Test
@DisplayName("특정 user 검색한다.")
@Order(3)
public void selectUserById() throws SQLException {
 String userid = "scoula";
 String sql = "select * from users where id = ?";
 try(PreparedStatement stmt = conn.prepareStatement(sql)){
 stmt.setString(1, userid);
 try(ResultSet rs = stmt.executeQuery()) {
 if(rs.next()) {
 System.out.println(rs.getString("name"));
 } else {
 throw new SQLException("scoula not found");
 }
 }
 }
}
```



# 1 JDBC 프로그래밍

## CrudTest.java

```
@Test
@DisplayName("특정 user 수정한다.")
@Order(4)
public void updateUser() throws SQLException {
 String userid = "scoula";
 String sql = "update users set name= ? where id = ?";
 try(PreparedStatement stmt = conn.prepareStatement(sql)){
 stmt.setString(1, "스콜라 수정");
 stmt.setString(2, userid);
 int count = stmt.executeUpdate();
 Assertions.assertEquals(1, count);
 }
}
```

# 1 JDBC 프로그래밍

## CrudTest.java

```
@Test
@DisplayName("지정한 사용자를 삭제한다.")
@Order(5)
public void deleteUser() throws SQLException {
 String userid = "scoula";
 String sql = "delete from users where id = ?";
 try(PreparedStatement stmt = conn.prepareStatement(sql)){
 stmt.setString(1, userid);
 int count = stmt.executeUpdate();
 Assertions.assertEquals(1, count);
 }
}
```

# 1 JDBC 프로그래밍

## ✓ VO 패턴

### ○ VO 객체

- Value Object
- 특정 테이블의 한 행을 매핑하는 클래스

클래스 정의 → 테이블

필드들 → 컬럼들

인스턴스 → 한 행

# 1 JDBC 프로그래밍

## UserVO.java

```
package org.scoula.jdbc_ex.domain;

import lombok.AllArgsConstructor;
import lombok.Data;
import lombok.NoArgsConstructor;

@Data
@NoArgsConstructor
@AllArgsConstructor
public class UserVO {
 private String id;
 private String password;
 private String name;
 private String role;
}
```

# 1 JDBC 프로그래밍

## ✓ DAO 패턴 적용

### ○ DAO 클래스

- Data Access Object
- 데이터베이스에 접근하여 실질적인 데이터베이스 연동 작업을 담당하는 클래스
- 테이블에 대한 CRUD 연산을 처리

### ○ 인터페이스 정의 후 구현 클래스 작성

 UserDao.java

```
package org.scoula.jdbc_ex.dao;

import org.scoula.jdbc_ex.domain.UserVO;

import java.sql.SQLException;
import java.util.List;
import java.util.Optional;

public interface UserDao {
 // 회원 등록
 int create(UserVO user) throws SQLException;

 // 회원 목록 조회
 List<UserVO> getList() throws SQLException;

 // 회원 정보 조회
 Optional<UserVO> get(String id) throws SQLException;

 // 회원 수정
 int update(UserVO user) throws SQLException;

 // 회원 삭제
 int delete(String id) throws SQLException;
}
```

## UserDaoImpl.java

```
package org.scoula.jdbc_ex.dao;

import org.scoula.jdbc_ex.common.JDBCUtil;
import org.scoula.jdbc_ex.domain.UserVO;

import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.util.ArrayList;
import java.util.List;
import java.util.Optional;

public class UserDaoImpl implements UserDao {
 Connection conn = JDBCUtil.getConnection();

 // USERS 테이블 관련 SQL 명령어
 private String USER_LIST = "select * from users";
 private String USER_GET = "select * from users where id = ?";
 private String USER_INSERT = "insert into users values(?, ?, ?, ?)";
 private String USER_UPDATE = "update users set name = ?, role = ? where id = ?";
 private String USER_DELETE = "delete from users where id = ?";
```

# 1 JDBC 프로그래밍

## UserDaoImpl.java

```
// 회원 등록
@Override
public int create(UserVO user) throws SQLException {
 try (PreparedStatement stmt = conn.prepareStatement(USER_INSERT)) {
 stmt.setString(1, user.getId());
 stmt.setString(2, user.getPassword());
 stmt.setString(3, user.getName());
 stmt.setString(4, user.getRole());
 return stmt.executeUpdate();
 }
}
```

```
private String USER_INSERT = "insert into users values(?, ?, ?, ?)";
```



## UserDaoImpl.java

```
private UserVO map(ResultSet rs) throws SQLException {
 UserVO user = new UserVO();
 user.setId(rs.getString("ID"));
 user.setPassword(rs.getString("PASSWORD"));
 user.setName(rs.getString("NAME"));
 user.setRole(rs.getString("ROLE"));
 return user;
}

// 회원 목록 조회
@Override
public List<UserVO> getList() throws SQLException{
 List<UserVO> userList = new ArrayList<UserVO>();
 Connection conn = JDBCUtil.getConnection();
 try (PreparedStatement stmt = conn.prepareStatement(USER_LIST);
 ResultSet rs = stmt.executeQuery()) {
 while(rs.next()) {
 UserVO user = map(rs);
 userList.add(user);
 }
 }
 return userList;
}
```

```
private String USER_LIST = "select * from users";
```

# 1 JDBC 프로그래밍

## UserDaoImpl.java

```
// 회원 정보 조회
@Override
public Optional<UserVO> get(String id) throws SQLException{
 try (PreparedStatement stmt = conn.prepareStatement(USER_GET)) {
 stmt.setString(1, id);
 try(ResultSet rs = stmt.executeQuery()) {
 if(rs.next()) {
 return Optional.of(map(rs));
 }
 }
 }
 return Optional.empty();
}
```

```
private String USER_GET = "select * from users where id = ?";
```

## 1 JDBC 프로그래밍

 UserDaoImpl.java

```
// 회원 수정
@Override
public int update(UserVO user) throws SQLException{
 Connection conn = JDBCUtil.getConnection();
 try (PreparedStatement stmt = conn.prepareStatement(USER_UPDATE)) {
 stmt.setString(1, user.getName());
 stmt.setString(2, user.getRole());
 stmt.setString(3, user.getId());
 return stmt.executeUpdate();
 }
}

// USERS 테이블 관련 CRUD 메소드
// 회원 삭제
@Override
public int delete(String id) throws SQLException{
 try(PreparedStatement stmt = conn.prepareStatement(USER_DELETE)) {
 stmt.setString(1, id);
 return stmt.executeUpdate();
 }
}
```

private String USER\_UPDATE = "update users set name = ?, role = ? where id = ?";

private String USER\_DELETE = "delete from users where id = ?";

# 1 JDBC 프로그래밍

## ✓ UserDao 테스트 클래스

```
public interface UserDao {
 // 회원 등록
 int create(UserVO user)

 // 회원 목록 조회
 List<UserVO> getList()

 // 회원 정보 조회
 Optional<UserVO> get(S

 // 회원 수정
 int update(UserVO user)

 // USERS 테이블 관련 CRU
 // 회원 삭제
 int delete(String id)
```

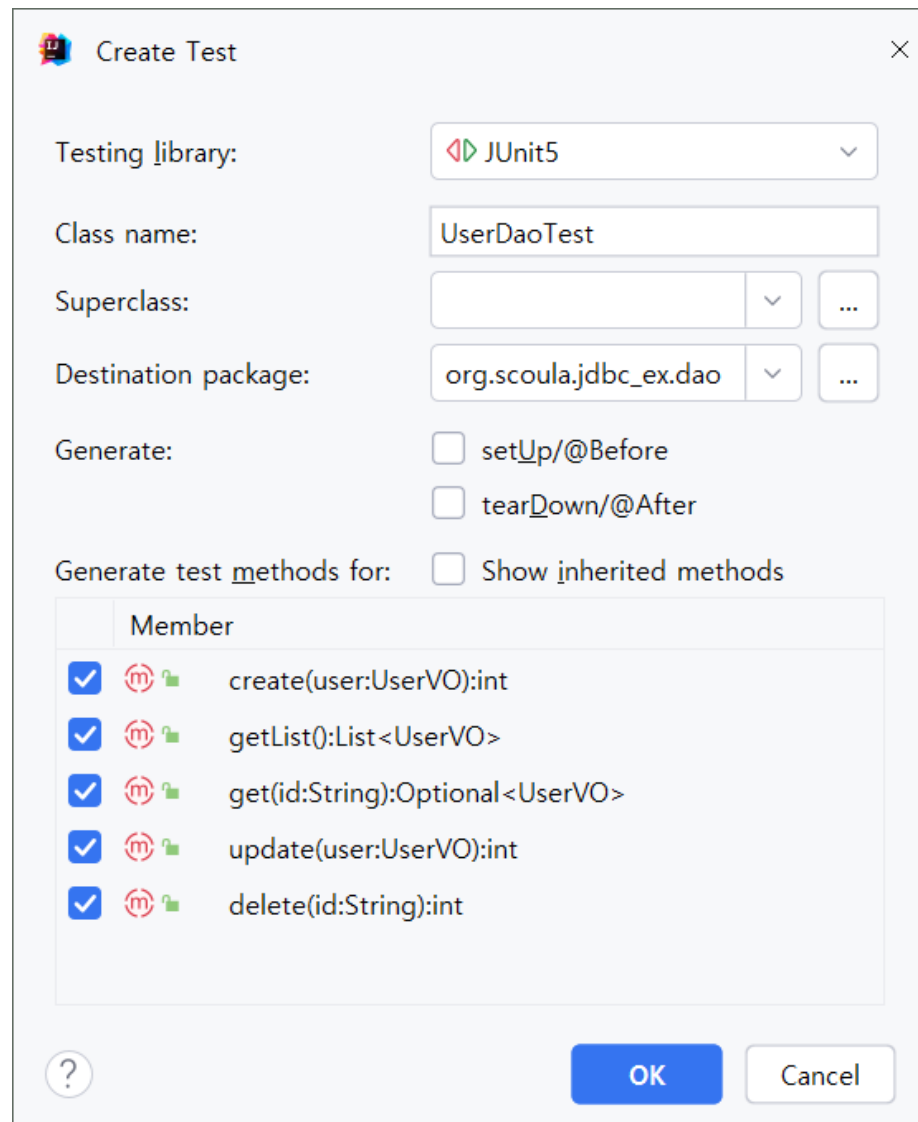
The screenshot shows an IDE with a context menu open over the `UserDao` interface. The menu includes options like 'Show Context Actions', 'Paste', 'Copy / Paste Special', 'Column Selection Mode', 'Find Usages', 'Go To', 'Folding', 'Analyze', 'Refactor', 'Generate...', and 'Open In'. The 'Go To' option is highlighted, and its submenu is open, displaying various navigation and development actions. The 'Test' option is highlighted in the submenu, with the keyboard shortcut `Ctrl+Shift+T` shown next to it.

**Choose Test for UserDao (0 found)** 📌

💡 Create New Test...

## 1 JDBC 프로그래밍

## ✓ UserDao 테스트 클래스



The image shows a 'Create Test' dialog box with the following fields and options:

- Testing library: JUnit5
- Class name: UserDaoTest
- Superclass: (empty)
- Destination package: org.scoula.jdbc\_ex.dao
- Generate: ☐ setUp/@Before, ☐ tearDown/@After
- Generate test methods for: ☐ Show inherited methods

A table of methods to be generated is shown below:

| Member                              |  |                                 |
|-------------------------------------|--|---------------------------------|
| <input checked="" type="checkbox"/> |  | create(user:UserVO):int         |
| <input checked="" type="checkbox"/> |  | getList():List<UserVO>          |
| <input checked="" type="checkbox"/> |  | get(id:String):Optional<UserVO> |
| <input checked="" type="checkbox"/> |  | update(user:UserVO):int         |
| <input checked="" type="checkbox"/> |  | delete(id:String):int           |

Buttons: ? (help), OK, Cancel

## 1 JDBC 프로그래밍

 UserDaoTest.java

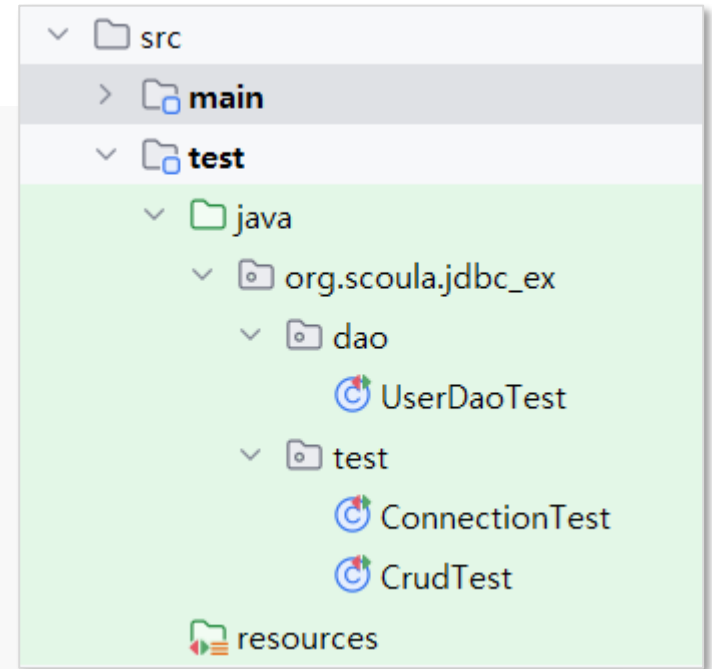
```
package org.scoula.jdbc_ex.dao;

import org.junit.jupiter.api.*;
import org.scoula.jdbc_ex.common.JDBCUtil;
import org.scoula.jdbc_ex.domain.UserVO;

import java.sql.SQLException;
import java.util.List;
import java.util.NoSuchElementException;

@TestMethodOrder(MethodOrderer.OrderAnnotation.class)
class UserDaoTest {
 UserDao dao = new UserDaoImpl();

 @AfterAll
 static void tearDown() {
 JDBCUtil.close();
 }
}
```



 UserDaoTest.java

```
@Test
@DisplayName("user를 등록합니다.")
@Order(1)
void create() throws SQLException {
 UserVO user = new UserVO("ssamz3", "ssamz123", "쌤즈", "ADMIN");
 int count = dao.create(user);
 Assertions.assertEquals(1, count);
}

@Test
@DisplayName("UserDao User 목록을 추출합니다.")
@Order(2)
void getList() throws SQLException {
 List<UserVO> list = dao.getList();
 for(UserVO vo: list) {
 System.out.println(vo);
 }
}
```

 UserDaoTest.java

```
@Test
@DisplayName("특정 user 1건을 추출합니다.")
@Order(3)
void get() throws SQLException {
 UserVO user = dao.get("ssamz3").orElseThrow(NoSuchElementException::new);
 Assertions.assertNotNull(user);
}

@Test
@DisplayName("user의 정보를 수정합니다.")
@Order(4)
void update() throws SQLException {
 UserVO user = dao.get("ssamz3").orElseThrow(NoSuchElementException::new);
 user.setName("쌤즈3");
 int count = dao.update(user);
 Assertions.assertEquals(1, count);
}

@Test
@DisplayName("user를 삭제합니다.")
@Order(5)
void delete() throws SQLException {
 int count = dao.delete("ssamz3");
 Assertions.assertEquals(1, count);
}
}
```